TEXORA · LECTURE SHEET

Fabric Spreading

Building the lay: ply after ply of fabric laid flat, aligned and tension-free to the marker's length — the requirements of a good lay, the packages fabric arrives in, the methods and machines that spread it, and the splice that rescues a lay when a roll ends or a fault appears.

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1. What spreading is

DEFINITION

Spreading is the laying of fabric in superimposed layers (plies), each ply the length of the marker, on the cutting table, to form the lay that will be cut in one operation. The number of plies equals the number of garments of each pattern piece required from that cutting (adjusted by the size mix of the marker).

Spreading looks like the simplest operation in the factory and is one of the most consequential. Every fault built into the lay — a stretched ply, a misaligned edge, a wrong face direction, a mixed shade — is multiplied by the cutting into every garment of the bundle. The requirements below are therefore not ideals; they are the conditions for the marker's plan to survive contact with real fabric.

2. Requirements of spreading

- **Correct ply count and length.** The lay must contain exactly the planned number of plies, each of marker length plus a small end allowance (a few centimetres at each end) for squaring and end loss.
- **Alignment.** All plies aligned at one end and along one edge, so every ply covers the whole marker; selvedges kept in a straight vertical line down the lay. A ply that falls short leaves the end pieces incomplete — hundreds of garments can be lost at one end of the table.
- **Freedom from tension.** Plies laid flat and relaxed, neither stretched nor slack. A stretched ply shrinks back after cutting and the panels come out small; knitted fabrics, which stretch easily and curl at the edges, must often be *relaxed* (left unrolled for hours) before spreading.
- **Correct face and nap direction.** Every ply faces the way the marker was planned: all face up, or face-to-face in pairs (§4), and pile/one-way fabrics all in one nap direction (Topic 4 constraints are executed here).

- **Shade control.** Plies from different dye lots are not mixed inside one garment's components: where a roll changes to a new shade lot, the change is made at a ply boundary and marked (ticketed), so bundles can be kept shade-pure for sewing.
- **On-grain laying.** The ply must lie square on the table — warp along the lay, weft at right angles — otherwise every piece in that ply is cut off-grain.
- **Static and cleanliness control.** Synthetic fabrics build static and cling or repel; the table and lay are kept clean so the bottom ply — the one that will show any mark — stays unmarked.

Fabric per lay = Ply length × Number of plies, where Ply length = Marker length + end allowances

Worked example. Marker 6.5 m, end allowance 4 cm at each end, 80 plies planned. Ply length = 6.5 + 0.08 = 6.58 m; fabric consumed = 6.58 × 80 = 526.4 m. The 6.4 m above the marker's exact 520 m is the spreading loss built into every lay — one reason end allowances are kept as small as alignment allows.

3. Fabric packages — how fabric arrives at the table

PACKAGE FORM	CONSEQUENCES FOR SPREADING
Open-width roll	The standard woven delivery: fabric wound flat on a tube. Fits travelling spreaders; one ply per pass.
Folded (book-folded) fabric	Fabric folded in zigzag layers in a carton or on a board. Suits face-to-face spreading of short lays; common for woollens and some domestic supply.
Tubular knitted fabric	Knit comes off the machine as a tube, and is spread as a doubled layer — two plies per pass — or slit open and spread open-width. Edge curl and width variation need care; relaxation before spreading is essential.
Cut pieces / remnants	Short lengths for samples and repair cuttings; spread by hand in miniature lays.

4. Spreading methods

4.1 By the way the plies face

MODE	HOW THE LAY IS BUILT	USED WHEN

Face-to-face (zigzag)	The spreader travels back and forth without cutting at the ends; the fabric folds at each end so plies alternate face-up/face-down in pairs.	Fastest mode; fabrics without nap direction; patterns cut as complete pairs. The classic mode for folded fabric.
Face one way	Every ply has the same face up; the fabric is cut at each end and the spreader returns (empty or carrying) to the start.	Nap, pile and one-way prints; asymmetric checks; whenever all pieces must point one way. Slower — the return trip produces nothing.

4.2 By the equipment used

- **Manual spreading** — two operators walk the roll (or draw fabric from a folded package) along the table, smoothing each ply by hand. Suitable for short lays, samples and delicate fabrics; slow and labour-heavy, but with the gentlest fabric handling.
- **Machine-assisted spreading** — a spreading machine carries the roll along the table while operators guide edges and ends; the common export-factory method.
- **Automatic spreading** — the machine is programmed with lay length and ply count, aligns edges by sensor, controls tension and cuts ends automatically; operators supervise and handle splices and shade changes.

5. Spreading machines

FEATURE	WHAT IT DOES
Roll cradle / feed	Carries the roll and unwinds it evenly, sometimes with powered unwinding for heavy rolls.
Travelling carriage	Moves the roll the length of the table on rails; speed matched to controlled, tension-free laying.
Edge alignment (photocell/sensor)	Senses the selvedge and shifts the carriage so all plies stack in one vertical edge line.
Tension control	Feeds fabric at the rate of travel so the ply is laid, not dragged — critical for knits.
End cutter and catcher	Cuts the ply squarely at the marker end and clamps it while the machine returns (face-one-way mode).
Ply counter and length setting	Stops the build at the programmed ply count and length.

Turntable	On some machines, rotates the roll 180° at each end so face-to-face lays can be built by a travelling machine.

The stationary alternative — fabric drawn from a fixed roll stand at the table end over a moving table or by hand — survives mainly in small units. Whatever the machine, its settings (length, count, tension) come from the marker plan of Topic 4: spreading executes what the marker decided.

6. Splices

DEFINITION

A splice is the overlapped join made in a ply during spreading where the fabric is cut and restarted — to remove a defective section, or where one roll ends and a new roll of the same fabric and shade begins. The two ends are overlapped for a short length (the splice lap, typically several centimetres, and more where a fault is cut out), kept perfectly on grain, and the position is marked.

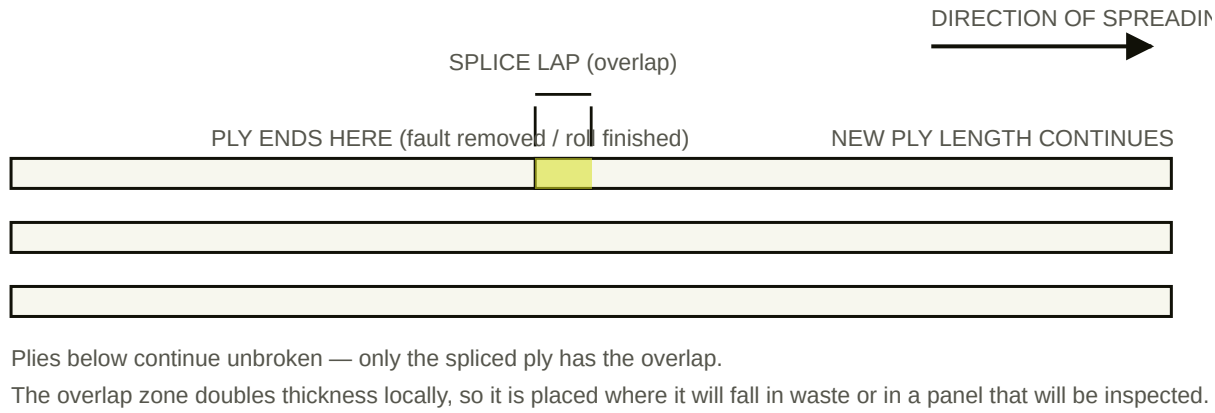


Figure 1 — A splice in one ply of a lay: the faulty or finished length ends, the new length overlaps it, and the position is marked for the cutting and inspection stages.

Rules of a good splice. (1) Cut out the whole fault with margin — a splice that leaves half a hole in a panel is worse than none. (2) Overlap squarely on grain, with the minimum lap that keeps both ends fully under the marker pieces above and below. (3) Place the splice, where the fabric allows, so the overlap falls between pattern pieces or inside a large, easily inspected panel. (4) Mark the splice position on the lay's edge and record it, so panels containing the overlap are checked after cutting. (5) Never splice across a shade change silently — a new lot begins a marked shade boundary, not a hidden splice (§2).

7. Glossary

TERM	MEANING

Ply	One layer of fabric in the lay; the lay's height is counted in plies.
Lay	The complete stack of plies spread for one cutting.
End allowance / end loss	The small extra length at each end of a ply for squaring, and the fabric it consumes.
Splice	The overlapped restart of a ply where fabric is faulty or a roll changes.
Relaxation	Resting fabric (especially knits) unrolled before spreading so it recovers its natural dimensions.
Shade lot	Fabric from one dyeing batch; components of one garment must come from one lot.

8. Key points and summary

- Spreading builds the lay: plies of marker length (+ end allowance), exactly counted, edge-aligned, tension-free, on grain, in the planned face/nap direction and shade order.
- Fabric arrives as open-width rolls, folded packages or tubular knits — each form dictates its spreading handling; knits need relaxation.
- Face-to-face spreading is fast and suits folded fabric and pair cutting; face-one-way is slower but compulsory for nap and one-way designs.
- Spreading machines contribute the cradle, carriage, edge sensors, tension control, end cutter and ply counter; automatic machines execute the marker plan under supervision.
- A splice overlaps a ply restart to remove faults or change rolls: cut fully, overlap on grain, place and mark it deliberately.

9. Practice

Answer in your own words; brief model answers follow.

1. Marker length 7.4 m, end allowance 5 cm at each end, 120 plies. Calculate the fabric consumed by the lay and the spreading loss over the exact marker length.
2. A knit fabric is spread straight off the roll under slight tension to “keep it neat”, and the cut panels are found 1.5% short. Explain the cause and the two steps that prevent it.
3. A velvet order must be spread face one way. State the cost of this mode in production terms and the garment fault it prevents.
4. During spreading, a hole is found in ply 47. Describe the correct splice procedure, step by step.

5. Why is a shade change between rolls handled as a marked boundary at a ply line instead of being spliced invisibly mid-ply?
6. List four things the spreading operator takes from the marker plan before starting a lay.

Answer key

1. Ply length = 7.4 + 0.10 = 7.5 m. Fabric = 7.5 × 120 = 900 m. Exact marker total = 7.4 × 120 = 888 m; spreading loss = 12 m.
2. The plies were laid stretched; after cutting, the fabric relaxed back and the panels shrank. Prevention: relax the fabric unrolled before spreading, and spread with tension control — feeding at travel speed, never dragging the ply tight.
3. Face-one-way costs the spreader's empty return trip (or end cut and repositioning) at every ply, so laying is slower and labour per lay higher. It prevents shade/direction mismatch: with nap, plies facing different ways would make garment panels reflect light differently and look like different colours.
4. Stop the lay at a ply boundary if possible; cut across the ply just beyond the hole with margin; restart the same fabric/shade overlapping the cut end by the splice lap, squared on grain; mark the splice on the lay edge and record it so the panel containing it is inspected after cutting.
5. Because all components of one garment must be one shade lot. A marked boundary at a ply line lets bundles be separated by lot; a hidden mid-ply splice would put two lots inside single panels and mixed garments that no later inspection can reliably untangle.
6. The lay length (marker length), ply count, face/nap mode for the build, and the size/set composition the marker represents (any of: shade instructions, splice/fault expectations, end allowances).

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